THERMAX LIMITED D-13, MIDC INDUSTRIAL AREA, R. D. AGA ROAD, CHINCHWAD, PUNE 411019. INDIA ☐ TEL.: (020) 27475941 (5 LINES) ☐ FAX : (020) 27475679 (D), 27472049 ☐ Website : www.thermaxindia.com ☐ IT PAN - AAACT3910D

Date: 17 March 2011

Cooling & Heating Division

THERMAX

ISO 9001 : 2000

Shell India Markets Private Limited
Powai Plaza, 401-403
4th Floor, Hiranandani Business Park
Powai, Mumbai – 400076

Kint Attn : Mr. Harish Koujalgi, Key Account Manager / Ameet Sevekar, Technical Manager

Dear Mr. Harish, Mr. Ameet

Sub: Approval for Shell Heat Transfer Oil S2 for Thermax Thermopac Heaters

Based on the enclosed technical parameters of **Shell Heat Transfer Oil S2**, we are pleased to confirm that **Shell Heat Transfer Oil S2** meets the specific technical requirements of Thermic Fluids for Thermopacs (Thermic Fluid Heaters) for the below mentioned medium temperature application.

Product	Maximum Bulk Operating Temperature
Shell Heat Transfer Oil S2	300°C

The above product, therefore, has recommendation for use in Thermax Thermopacs (Thermic Fluid Heaters) in India, Nepal and Bhutan.

This recommendation of Thermic Fluid will be included in the subsequent revision of the Operations & Maintenance manual of Thermopac. This approval supersedes all our previous approvals of Medium Temperature Thermic Fluids. For maximum Bulk Operating Temperature exceeding 280°C at TF pump inlet, we recommend to use Mechanical seals for Circulation Pump (for pumps which do not have built in mechanical sealing system with air cooled bearings arrangement or magnetic coupling arrangements) and Cold Sealpot with Expansion Tank. It is expected that Thermic Fluid system design is done as per Thermax recommendations and proper engineering practices.

Yours truly,

For **Thermax Limited**

RCPK
21/3/11

Sanjay Kumar Misri
Head – Marketing & Product Management

R.C. Patilkulkarni
Head – Product Engineering

Encl: Technical Data Sheet for Shell Heat Transfer Oil S2

Previous Name: **Shell Thermia B**

Shell **Heat Transfer Oil *S2***

High Performance Heat transfer fluid

- ***RELIABLE PERFORMANCE***

Shell Heat Transfer Oil S2 is based on carefully selected, highly refined mineral oils chosen for their ability to provide superior performance in indirect closed fluid heat transfer systems.

Applications

- ***Enclosed circulated heat transfer systems***
  for industrial applications such as process industry, chemical plants, textile producers etc. and in household equipment such as oil filled radiators. Shell Heat Transfer Oil S2 can be used in high temperature continuous heat transfer equipment with the following application limits:

Shell Heat Transfer Oil S2	
Max. film temperature	340°C
Max. bulk temperature	320°C

Performance Features and Benefits

- ***Extended maintenance intervals***
  Shell Heat Transfer Oil S2 is based on carefully selected highly refined mineral oils and resists oil cracking, oxidation and thickening. This provides extended oil life, provided efficient fluid heating and good pump circulation is ensured, such that film temperatures on the heater surface do not exceed the limits above.
- ***System efficiency***
  Low viscosity enables excellent fluidity and heat transfer over a wide temperature range.
  Shell Heat Transfer Oil S2 also has a low vapour pressure so resists cracking. This minimises the formation of volatile decomposition products; these would require recovery via expansion chamber and condensate collector
- ***Wear protection***
  Shell Heat Transfer Oil S2 is non-corrosive and has high solvency – this reduces deposit formation by holding oxidation products in solution and keeping internal surfaces of heat exchangers clean.

Specification and Approvals

Classified as ISO 6743-12 Family Q
Meets typically DIN 51522 requirements

Advice

The life of Shell Heat Transfer Oil S2 depends on the design and usage of the system. If the system is well designed and not subjected to abnormal workloads, the life can be for many years.

It is important to monitor oil condition regularly as rates of change in physical characteristics are more significant than actual values. The properties that should be monitored are viscosity, acidity, flash point (open and closed) and insolubles content.

Advice on applications not covered in this leaflet may be obtained from your Shell representative.

Health and Safety

Guidance on Health and Safety is available on the appropriate Material Safety Data Sheet which can be obtained from your Shell representative.

Protect the environment

Take used oil to an authorised collection point. Do not discharge into drains, soil or water.

SANJAY MISRI
17/3/11

21/3/11
R.C. Pabikulkarni

Typical Physical Characteristics

Density at 15 °C	kg/m3	ISO 12185	866
Flash Point PMCC	°C	ISO 2719	210
Flash Point COC	°C	ISO 2592	220
Fire Point COC	°C	ISO 2592	255
Pour Point	°C	ISO 3016	-12
Kinematic Viscosity		ISO 3104	
at 0 °C	mm2/s		151
at 40 °C	mm2/s		25
at 100 °C	mm2/s		4.7
at 200 °C	mm2/s		1.1
Initial Boiling Point	°C	ASTM D 2887	355
Autoignition Temperature	°C	DIN 51794	360
Neutralisation Value	mgKOH/g	ASTM D974	< 0.05
Ash (Oxid)	%m/m	ISO 6245	< 0.01
Carbon Residue (Conradson)	%m/m	ISO 10370	0.02
Copper Corrosion (3h/100°C)		ISO 2160	class 1

These characteristics are typical of current production. Whilst future production will conform to Shell's specification, variations in these characteristics may occur.

Typical Design Data

Temperature	°C		0	20	40	100	150	200	250	300	340
Density	kg/m3		876	863	850	811	778	746	713	681	655
Specific Heat Capacity	kJ/kg*K		1.809	1.882	1.954	2.173	2.355	2.538	2.72	2.902	3.048
Thermal Conductivity	W/m*K		0.136	0.134	0.133	0.128	0.125	0.121	0.118	0.114	0.111
Prandtl No.			3375	919	375	69	32	20	14	11	9

(SANJAY MISRI)
17/3/11

RCPK
21/3/11
R.C. Pabikulkarni